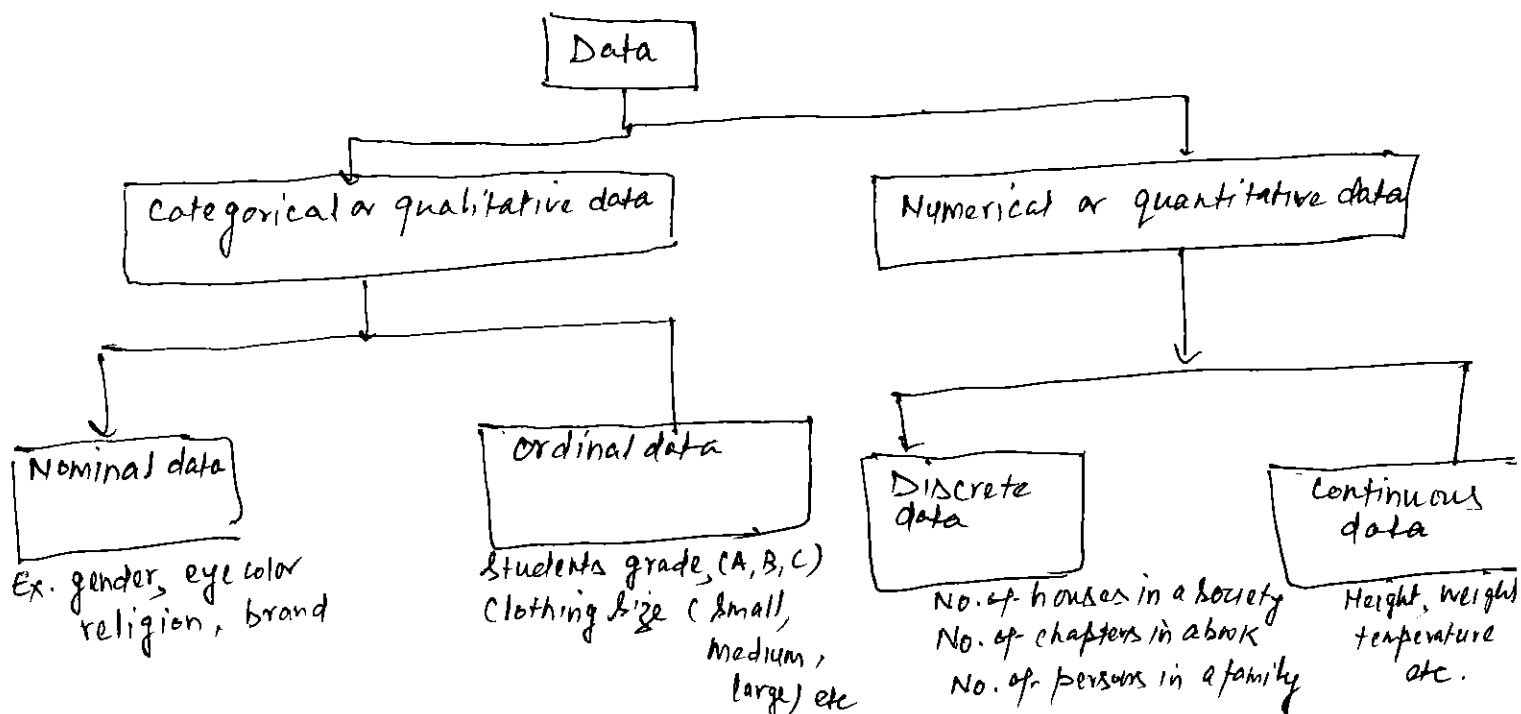


Def<sup>n</sup> of Statistics :

- 1) As a plural noun, is used to mean numerical data arising in any sphere of human experience.
- 2) As a singular noun, statistics is a name ~~of~~ for the body of scientific methods (the statistical methods) which are meant for the collection, analysis and interpretation of numerical data.
- 3) By Croxton and Cowden : "statistics may be defined as "the science which deal with the collection, analysis and interpretation of numerical data"
- 3) statistics is the science of ~~the~~ data.
- 4) statistics is the science of decision making.

Data : Numerical representation of facts

Types of Data

## Concept of frequency distribution

Purpose : To summarize the data.

Two types of frequency distn

I) Ungrouped frequency distribution

| X        | f        |
|----------|----------|
| $x_1$    | $f_1$    |
| $x_2$    | $f_2$    |
| $\vdots$ | $\vdots$ |
| $x_n$    | $f_n$    |

$N$

$$N = \sum_{i=1}^n f_i$$

| X | f |
|---|---|
| 4 | 3 |
| 5 | 2 |
| 6 | 5 |
| 7 | 3 |

II) Grouped type frequency distribution

| Class intervals | frequency (f) | Tally marks |
|-----------------|---------------|-------------|
| 0 — 9           | 5             |             |
| 10 — 19         | 10            |             |
| 20 — 29         | 15            |             |
| 30 — 39         | 8             |             |

$$N = 38$$

| class-intervals | frequency |
|-----------------|-----------|
| 0 — 10          | 5         |
| 10 — 20         | 10        |
| 20 — 30         | 15        |
| 30 — 40         | 11        |
| 40 — 50         | 7         |

$$N = 48$$

No. of classes should not be too small or too large

it should be preferably between 5 and 15

Struges gave a formula: approximate no. of classes should be

$$K = 1 + 3.322 \log_{10} N : \text{ where } N = \sum_{i=1}^n f_i$$

$$R = H - L$$

$$\text{Class width } W \approx \frac{R}{K} \text{ (nearest integer)}$$

How to make discrete class intervals continuous types  
 If  $d$  is the gap between the upper limit of any class and lower limit of the succeeding class, the class boundaries for any class are then given by

$$\text{Upper class boundary} = \text{upper class limit} + \frac{d}{2}$$

$$\text{Lower class boundary} = \text{Lower class limit} - \frac{d}{2}$$

Ex

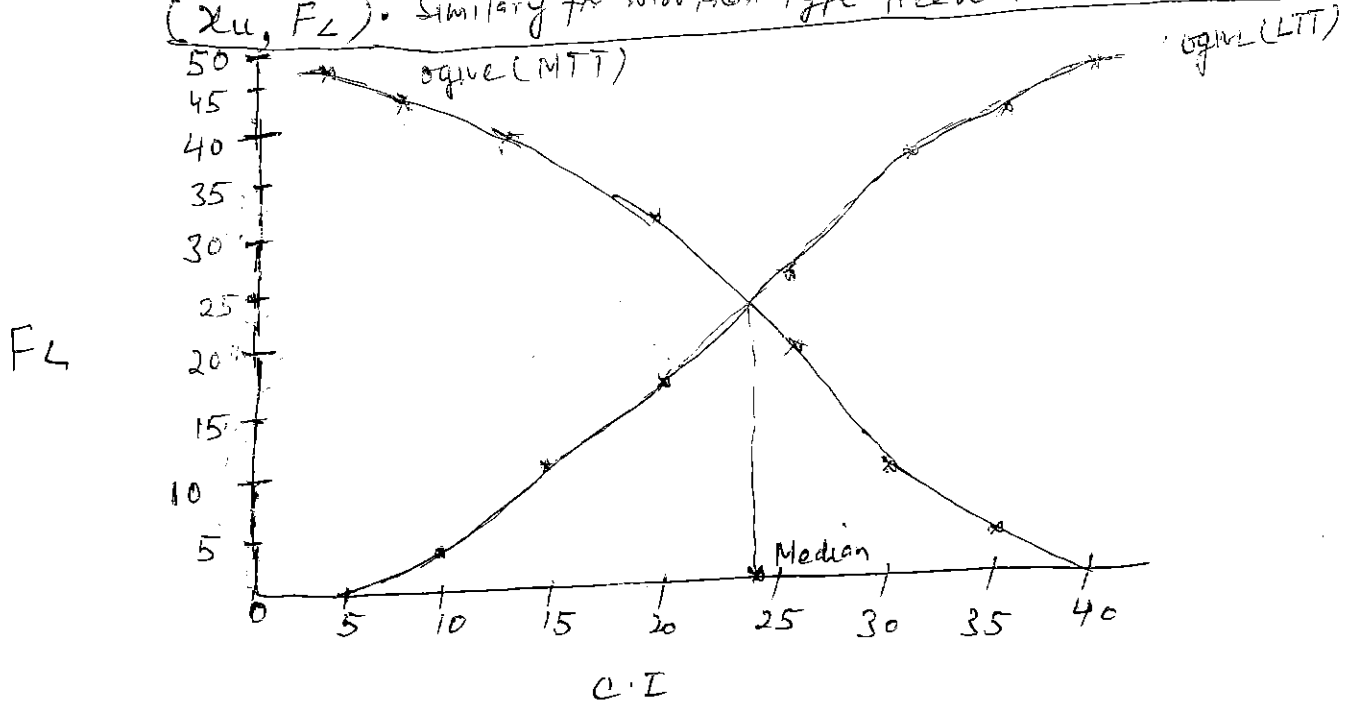
| Discrete C.I | Continuous C.I |
|--------------|----------------|
| 15 - 19      | 14.5 - 19.5    |
| 20 - 24      | 19.5 - 24.5    |
| 25 - 29      | 24.5 - 29.5    |
| 30 - 34      | 29.5 - 34.5    |
| 35 - 39      | 34.5 - 39.5    |

Various columns in a frequency distribution table

| C.I     | freq.<br>$f$ | Rel. freq.<br>$f/N$ | Cumulative Freq                |                         |
|---------|--------------|---------------------|--------------------------------|-------------------------|
|         |              |                     | Less than type<br>$F_L$ or $C$ | More than type<br>$F_7$ |
| 5 - 10  | 3            | 3/48                | 3                              | 48                      |
| 10 - 15 | 5            | 5/48                | 8                              | 45                      |
| 15 - 20 | 8            | 8/48                | 16                             | 40                      |
| 20 - 25 | 12           | 12/48               | 28                             | 32                      |
| 25 - 30 | 10           | 10/48               | 38                             | 20                      |
| 30 - 35 | 6            | 6/48                | 44                             | 10                      |
| 35 - 40 | 4            | 4/48                | 48                             | 4                       |
|         | 48           | 1.00                |                                |                         |

ogive less than type or Cumulative

frequency curve less than type. Free hand curve between  $(\Sigma u, F_L)$ . Similar for more than type free hand curve between  $(\Sigma L, F_U)$



### Properties of Data

- i) Central tendency      (ii) Dispersion  
(iii) Skewness      (iv) Kurtosis

Central tendency: is the phenomenon or characteristics of data to concentrate around a central value.

Dispersion tells how the observations are scattered or dispersed from a central value.

Skewness tells us ~~how~~ whether the data is symmetrical or asymmetrical

Kurtosis tells us whether the frequency curve is peaked or flat

Requisition for an ideal measure of central tendency

- i) It should be rigidly defined (ii) It should be readily comprehensible and easy to calculate (iii) It should be based upon all the observations (iv) It should be suitable for further use (v) It should be least affected by the fluctuations of sampling.  
(vi) Should not be much affected by the extreme observations.  
(Not by Prof Yule, added on latter)

Measures of Central tendency

- i) Mean      ii) Median      iii) Mode

i) Mean

- (a) Airthmetic Mean (AM)      (b) Geometric Mean (GM)      (c) Harmonic Mean (HM)

Airthmetic Mean (AM)

For raw data :  $x_1, x_2, \dots, x_n$

A.M.       $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

For frequency data (ungrouped type)

| X.    | f     | fx        |
|-------|-------|-----------|
| $x_1$ | $f_1$ |           |
| $x_2$ | $f_2$ |           |
| $x_n$ | $f_n$ |           |
|       | N     | $\sum fx$ |

A.M.       $\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i$

Where  $N = \sum_{i=1}^n f_i$

# FN Grouped type

| C.I   | x  | f  | fx  |
|-------|----|----|-----|
| 0-10  | 5  | 3  | 15  |
| 10-20 | 15 | 4  | 60  |
| 20-30 | 25 | 5  | 125 |
| 30-40 | 35 | 2  | 70  |
|       |    | 14 | 270 |

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i$$

$$= \frac{270}{14} = 19.29$$

## Effect of change of origin and scale on AM

| x | f |
|---|---|
|---|---|

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i$$

$$N = \sum_{i=1}^n f_i$$

$$U = \frac{x-A}{h} ; \text{ where } A + h \text{ are suitably chosen constants}$$

$$x = A + hU$$

$$x_i = A + hU_i$$

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i (A + hU_i) = A + h\bar{U}$$

$$\text{Where } \bar{U} = \frac{1}{N} \sum_{i=1}^n f_i U_i$$

$$\boxed{\bar{x} = A + h\bar{U}}$$

i) If  $A = 0$  (change of scale only)

$$\bar{x} = h\bar{U}$$

ii) If  $h = 1$

$$\bar{x} = A + \bar{U} \quad (\text{change of origin only})$$

Properties of AM : 1) Sum of deviations of a set of observations is zero, if deviations are taken from their AM

i) For raw data  $\sum_{i=1}^n (x_i - \bar{x}) = 0$  ; it is obvious

$$\sum_{i=1}^n x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0$$

ii) For freq<sup>d</sup> data  $\sum_{i=1}^n f_i (x_i - \bar{x}) = 0 \Rightarrow \sum_{i=1}^n f_i x_i - N\bar{x} = N\bar{x} - N\bar{x} = 0$

ii) Sum of squares of deviations of a set of observations is minimum if the deviations are taken from the mean.

For raw data  $Z = \sum_{i=1}^n (x_i - A)^2$

$$\frac{\partial Z}{\partial A} = 2 \sum_{i=1}^n (x_i - A)(-1) = 0$$

gives  $A = \frac{1}{n} \sum x_i = \bar{x}$

$$\frac{\partial^2 Z}{\partial A^2} = 2N > 0, \text{ Hence, } Z \text{ is minimum at } A = \bar{x}$$

For freq<sup>n</sup> data  $Z = \sum_{i=1}^n f_i (x_i - A)^2$

$$\frac{\partial Z}{\partial A} = 2 \sum_{i=1}^n f_i (x_i - A)(-1) = 0$$

gives  $A = \frac{1}{N} \sum f_i x_i = \bar{x}$

$$\frac{\partial^2 Z}{\partial A^2} = 2N > 0, \text{ Hence, } Z \text{ is minimum at } A = \bar{x}$$

iii) Mean of Composite Series

Diagram illustrating the mean of composite series:

- Group 1: Frequency  $n_1$ , values  $x_{11}, x_{12}, \dots, x_{1n_1}$ , mean  $\bar{x}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} x_{1j}$
- Group 2: Frequency  $n_2$ , values  $x_{21}, x_{22}, \dots, x_{2n_2}$ , mean  $\bar{x}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} x_{2j}$
- Group k: Frequency  $n_k$ , values  $x_{k1}, x_{k2}, \dots, x_{kn_k}$ , mean  $\bar{x}_k = \frac{1}{n_k} \sum_{j=1}^{n_k} x_{kj}$
- Overall Mean:  $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + \dots + n_k \bar{x}_k}{(n_1 + n_2 + \dots + n_k)}$

This property also proves that AM is suitable for further mathematical use.

## Geometric Mean

raw data  $x_1, x_2, \dots, x_n$

$$G = (x_1 x_2 \dots x_n)^{\frac{1}{n}}$$

$$\begin{aligned} \log G &= \frac{1}{n} [\log x_1 + \log x_2 + \dots + \log x_n] \\ &= \frac{1}{n} \sum_{i=1}^n \log x_i \end{aligned}$$

$$G = \text{Antilog} \left[ \frac{1}{n} \sum_{i=1}^n \log x_i \right]$$

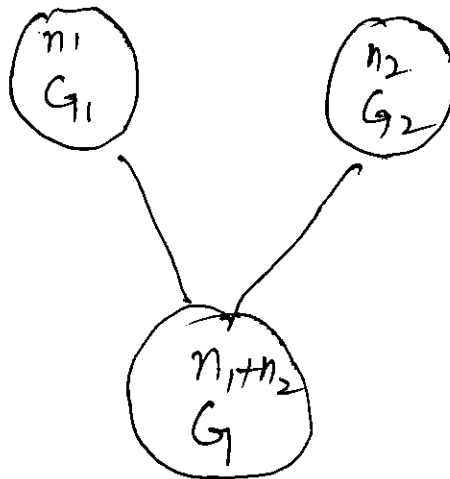
$x/f$

$$G = (x_1^{f_1} x_2^{f_2} \dots x_n^{f_n})^{\frac{1}{N}}$$

$$N = \sum_{i=1}^n f_i$$

$$G = \text{Antilog} \left[ \frac{1}{N} \sum_{i=1}^n f_i \log x_i \right]$$

## G.M of Composite Series



$$\log G = \frac{1}{n_1+n_2} [n_1 \log G_1 + n_2 \log G_2]$$

## Harmonic Mean

For raw data

$$H = \frac{1}{\frac{1}{n} \sum_{i=1}^n \left( \frac{1}{x_i} \right)}$$

For freq<sup>n</sup> data  $x/f$

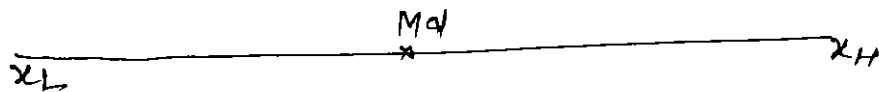
$$H = \frac{1}{\frac{1}{N} \sum_{i=1}^n \left( \frac{f_i}{x_i} \right)}, \quad N = \sum_{i=1}^n f_i$$



## Median :

Median is the middle most observations, which divides the whole set of data into two equal halves.

It is also known as positional measure and applicable for qualitative data and wage distribution.



For raw data:  $x_1 \ x_2 \ \dots \ x_n$

Step 1 arrange the data into ascending/descending order

If  $n$  is odd  $\frac{n+1}{2}$ th observation is median

drawback  $\leftarrow$  If  $n$  is even, Average of  $\frac{n}{2}$ th and  $\frac{n}{2}+1$ th observations

For ungrouped freq<sup>n</sup> dist<sup>n</sup>

Calculate  $\frac{N}{2}$ ,

If  $\frac{N}{2} \leq F_i$  (CFLTT just greater than or equal to  $\frac{N}{2}$ )

$Md = x_i$

CFLTT

| $x$   | $f$   | $C \text{ or } F_L$ |
|-------|-------|---------------------|
| $x_1$ | $f_1$ | $F_1$               |
| $x_2$ | $f_2$ | $F_2$               |
| $x_n$ | $f_n$ | $F_n$               |

Ex

$$\frac{N}{2} = \frac{28}{2} = 14$$

In the col<sup>n</sup> of CFLTT 19 is just greater than 14, therefore, median is 5.

$Md = 5$

| $x$ | $f$ | $F_L$ |
|-----|-----|-------|
| 2   | 1   | 1     |
| 3   | 3   | 4     |
| 4   | 5   | 9     |
| 5   | 10  | 19    |
| 6   | 4   | 23    |
| 7   | 3   | 26    |
| 8   | 2   | 28    |

$N = 28$

Median is an insensitive measure

2   4   5   8   10    $\rightarrow$     $Md = 5$

1   2   4   5   8   10   12    $\rightarrow$     $Md = 5$  (Added both side)

4   5   8    $\rightarrow$     $Md = 5$  (deleted both side)

## For grouped frequency dist<sup>n</sup>

(For Cont. Class Interval)

$$Md = l + \frac{h}{f} \left( \frac{N}{2} - C \right) ;$$

$l$ : lower limit of median class

$h$ : magnitude/class interval of the median class

$f$ : frequency of the median class

$C$ : CF(LTT) of the class preceding to the median class

$$N = \sum_{i=1}^n f_i$$

### Median class

Median class is the class corresponding to the c.f just greater than or equal to  $\frac{N}{2}$ .

Ex

| C.I     | f  | CF(LTT) |
|---------|----|---------|
| 0 — 10  | 5  | 5       |
| 10 — 20 | 8  | 13      |
| 20 — 30 | 12 | 25      |
| 30 — 40 | 10 | 35      |
| 40 — 50 | 6  | 41      |

$N = 41$

$$\frac{N}{2} = \frac{41}{2} = 20.5$$

Md class

$$Md = 20 + \frac{10}{12} (20.5 - 13)$$

$$= 20 + 0.83 (7.5)$$

$$= 20 + 6.23 = 26.23$$

Merits to be explained  
Demerits to be explained

Mode: Mode is the most frequent observation in the data. Applicable in business field

- i) For raw data: observations repeating maximum times
- ii) ~~Un~~grouped freq<sup>n</sup> dist<sup>n</sup>: For symmetrical dist<sup>n</sup> mode is corresponding to the highest frequency.

Note: If highest freq<sup>n</sup> is ~~repeating~~ occurring in beginning or at the end, the method of grouping is required.

## For grouped (Cont. Class Interval) freq<sup>n</sup> dist<sup>n</sup>

$$Mode (Mo) = l + \frac{h(f_1 - f_0)}{(2f_1 - f_0 - f_2)} ; \text{ Modal class: Corresponding to the highest freq.}$$

where  $l$ : lower limit of modal class,  $h$ : class width of the modal class  
 $f_1$ : freq<sup>n</sup> of modal class,  $f_0$  and  $f_2$  are frequencies of the classes preceding and succeeding the modal class

Ex.

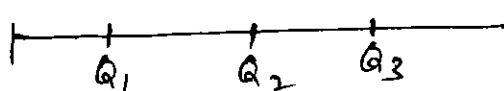
| C.I | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|-----|------|-------|-------|-------|-------|-------|-------|-------|
| f   | 5    | 8     | 7     | 12    | 28    | 20    | 10    | 8     |

Highest freq<sup>n</sup> is 28, hence the modal class is 40-50  
 Mode =  $40 + \frac{10(28-12)}{2 \times 28 - 12 - 20} = 40 + 6.67 = 46.67$ .

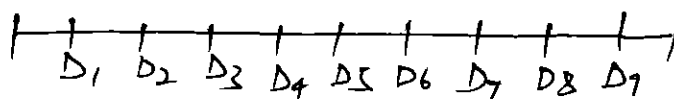
Merits + Demerits of mode to be explained.

Partition values (Quantiles)

Quartiles ( $Q_1, Q_2, Q_3$ ): Divides whole set of data into four equal parts.



Deciles ( $D_1, D_2, \dots, D_9$ ): Divides whole set of data into ten equal parts



Percentiles ( $P_1, P_2, \dots, P_{99}$ ): Divides whole set of data into 100 equal parts

Calculation is similar to that of median

$$Q_i = d + \frac{h}{f} \left( \frac{iN}{4} - C \right); (i=1, 2, 3)$$

$$D_i = d + \frac{h}{f} \left( \frac{iN}{10} - C \right); (i=1, 2, \dots, 9)$$

$$P_i = d + \frac{h}{f} \left( \frac{iN}{100} - C \right); (i=1, 2, \dots, 99)$$

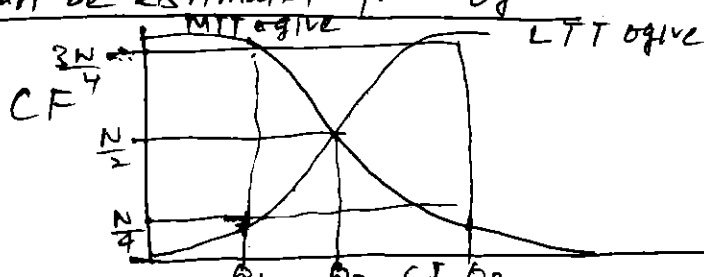
Quantiles can be estimated from ogives

In LTT ogive

$$Q_1 \rightarrow \frac{N}{4}$$

$$Q_2 \rightarrow \frac{2N}{4} = \frac{N}{2}$$

$$Q_3 \rightarrow \frac{3N}{4}$$



In MTT ogive

$$Q_1 = \frac{3N}{4}$$

$$Q_2 = \frac{N}{2}$$

$$Q_3 = \frac{N}{4}$$